

Name : _____

Mark : _____

Date: / /2023

Period: 2 hours Pages: 4



BENG04 CHEMISTRY REEXAM
2022-2023

REPUBLIC OF YEMEN
Science & Technology University
(Aden)
Faculty of Engineering
Civil Engineer Department
Examiner: Dr. Adel A. M. Saeed

Write all the questions below (it is allowed to use a calculator)

Q.1 Choose the correct answers (10 marks):

(a) In thermodynamics, work has a broader meaning that includes _____.

(1) mechanical work, (2) kinetic work, (3) electrical work, (4) all are wrong

(b) The SI unit for energy is _____.

(1) the calorie (cal), (2) the Joule (J), (3) the kelvin (k), (4) all are right

(c) A physical change to convert gas to liquid is _____.

(1) sublimation, (2) freezing, (3) evaporation, (4) condensation

(d) Chemical reactions have several types such as _____.

(1) decomposition, (2) redox reactions, (3) 1 & 2, (4) all are wrong

(e) The chemical composition of rust is _____.

(1) FeO, (2) FeCO₃, (3) Fe₂O₃.XH₂O, (4) FeO.XH₂O

Q.2 Complete the table below (10 marks):

Cation	Anion	Chemical formula	Systematic name
Al ³⁺	CO ₃ ²⁻		
NH ₄ ⁺	SO ₄ ²⁻		
Na ⁺	ClO ₃ ⁻		
H ⁺	IO ₄ ⁻		
K ⁺	S ²⁻		

Q.3 Give the order of the following (10 marks):

(a) Atomic radius: ¹¹Na, ¹⁷Cl, ³Li, ⁸O

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(b) Electronegativity: ¹²Mg, ³Li, ⁹F, ⁸O

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(c) *Electron Configurations:* $_{25}\text{Mn}$, $_{7}\text{N}$, $_{10}\text{Ne}$, $_{16}\text{S}$

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(d) *The entropy (S) of the molecules:* HI , HBr , HF , HCl

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(e) *The volumes:* $20,000\ \mu\text{L}$, $2.00 \times 10^3\ \text{mL}$, $200\ \text{cm}^3$, $200,000\ \text{nL}$

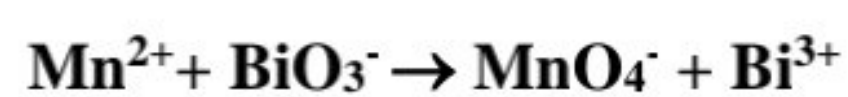
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Q.4

CHOOSE ONLY TWO

Perform the following calculations (10 marks each)

(1) Complete and balance the following reactions:



(2) What is the molality of a solution prepared by mixing 10 mL of $\text{C}_2\text{H}_6\text{O}_2$ ($d = 1.2$ g/mL; M.wt = 62) with 300 g of H_2O ?

(3) How many atoms, moles and mass of each element in the reaction



(A.wt: Cr= 52 , O= 16, H=1, Cl=35.5)

Good Luck